

solar_installation_report

May 9, 2026

1 Project: Solar Energy Installation Analysis

Solar energy has become one of the fastest-growing renewable energy sources in Australia. Many households and businesses are installing solar panel systems to reduce electricity costs and lower carbon emissions. This project analyses solar installation and solar capacity trends across Australian states and territories between 2011 and 2025.

Objectives:

- Examine trends in solar installations over time,
- Compare solar adoption between states and territories,
- Identify growth patterns in renewable energy usage across Australia.
- **Workflow:** Load → Explore → Clean (filter data) → Aggregation (by region) → Visualization → Insights

Data Sources and References This analysis uses sensitive internal datasets provided by Australian Renewable Energy Agency (ARENA). ARENA is interested in understanding the factors influencing solar energy adopt on across different states and territories in Australia. All data and copyright being used belong to ARENA.

All data remains the property of ARENA and is restricted to analytical and reporting purposes within the organisation. The data set includes postcode-level solar installation records, state/territory identifiers, and population estimates. This datasets are also provided by the Australian Bureau of Statistics (ABS).

Links

- Both `sgu-solar-capacity-2011-to-present-and-total` and `sgu-solar-installations-2011-to-present-and-total`, Provided in starter pack for this assignment by NMTAFE.
- <https://www.abs.gov.au/statistics/people/population/national-state-and-territory-population>
- <https://www.abs.gov.au/statistics/people/population/national-state-and-territory-population/sep-2025>
- <https://www.abs.gov.au/statistics/people/population/national-state-and-territory-population/latest-release>

1.0.1 Methodology

Python was used for the analysis with the following libraries:

- Pandas for data manipulation,
- NumPy for numerical operations,

- Matplotlib and Seaborn for data visualization.

The analysis involved:

1. loading the datasets,
2. cleaning and transforming the data,
3. grouping records by year and state,
4. visualizing solar trends across Australia.

```
[2]: # install packages for said project
%pip install pandas
%pip install numpy
%pip install seaborn
%pip install matplotlib
```

```
Requirement already satisfied: pandas in
C:\Users\kabyw\source\repos\MyJupyter\.venv\Lib\site-packages (3.0.2)
Requirement already satisfied: numpy>=1.26.0 in
C:\Users\kabyw\source\repos\MyJupyter\.venv\Lib\site-packages (from pandas)
(2.4.4)
Requirement already satisfied: python-dateutil>=2.8.2 in
C:\Users\kabyw\source\repos\MyJupyter\.venv\Lib\site-packages (from pandas)
(2.9.0.post0)
Requirement already satisfied: tzdata in
C:\Users\kabyw\source\repos\MyJupyter\.venv\Lib\site-packages (from pandas)
(2026.1)
Requirement already satisfied: six>=1.5 in
C:\Users\kabyw\source\repos\MyJupyter\.venv\Lib\site-packages (from python-
dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
Requirement already satisfied: numpy in
C:\Users\kabyw\source\repos\MyJupyter\.venv\Lib\site-packages (2.4.4)
Note: you may need to restart the kernel to use updated packages.
```

```
[1]: # Step 1: Load the data
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load CSV
solar_capacity_df = pd.read_csv('sgu-solar-capacity-2011-to-present-and-totals.
↪csv')
print(f"Loaded {len(solar_capacity_df):,} rows × {len(solar_capacity_df.
↪columns)} columns")
solar_capacity_df.head()
```

Loaded 2,810 rows × 178 columns

```

[1]: Small Unit Postcode Historic Total Rated Power Output In kW (2001 - 2010) \
0      0      4.46
1      200     0.08
2      800    34.74
3      801     3.44
4      803     0.00

Jan 2011 - Rated Power Output In kW Feb 2011 - Rated Power Output In kW \
0      0.0      0.0
1      0.0      0.0
2      0.0      0.0
3      0.0      0.0
4      0.0      0.0

Mar 2011 - Rated Power Output In kW Apr 2011 - Rated Power Output In kW \
0      0.0      0.0
1      0.0      0.0
2      0.0      0.0
3      0.0      0.0
4      0.0      0.0

May 2011 - Rated Power Output In kW Jun 2011 - Rated Power Output In kW \
0      0.00      0.0
1      0.00      0.0
2      2.22      0.0
3      2.02      0.0
4      0.00      0.0

Jul 2011 - Rated Power Output In kW Aug 2011 - Rated Power Output In kW \
0      0.00      0.0
1      0.00      0.0
2      3.04      0.0
3      0.00      0.0
4      0.00      0.0

... Nov 2024 - Rated Power Output In kW \
0 ...      0.000
1 ...      0.000
2 ...     123.045
3 ...      0.000
4 ...      0.000

Dec 2024 - Rated Power Output In kW Jan 2025 - Rated Power Output In kW \
0      0.0      0.0
1      0.0      0.0
2      0.0      0.0
3      0.0      0.0

```

4	0.0	0.0
	Feb 2025 - Rated Power Output In kW	Mar 2025 - Rated Power Output In kW \
0	0.0	0.0
1	0.0	0.0
2	0.0	274.5
3	0.0	0.0
4	0.0	0.0
	Apr 2025 - Rated Power Output In kW	May 2025 - Rated Power Output In kW \
0	0.00	0.0
1	0.00	0.0
2	99.88	0.0
3	0.00	0.0
4	0.00	0.0
	Jun 2025 - Rated Power Output In kW	Jul 2025 - Rated Power Output In kW \
0	0.00	0.0
1	0.00	0.0
2	164.23	0.0
3	0.00	0.0
4	0.00	0.0
	Total Rated Power Output In kW	
0	4.460	
1	0.080	
2	4928.175	
3	386.598	
4	11.685	

[5 rows x 178 columns]

1.1 Dataset Description

Two datasets were used in this project:

1. Solar Capacity Dataset

- Measures solar power output in kilowatts (kW)
- Contains monthly data from 2011–2025
- Includes postcode-based records across Australia

2. Solar Installation Dataset

- Measures the number of solar installations
- Covers the same time period and postcode regions

```
[2]: solar_capacity_df.info()
solar_capacity_df.describe()
```

```

<class 'pandas.DataFrame'>
RangeIndex: 2810 entries, 0 to 2809
Columns: 178 entries, Small Unit Postcode to Total Rated Power Output In kW
dtypes: float64(177), int64(1)
memory usage: 3.8 MB

```

```

[2]:      Small Unit Postcode  \
count      2810.000000
mean      4168.266548
std       1592.968085
min         0.000000
25%       2845.250000
50%       3929.500000
75%       5340.750000
max       9729.000000

      Historic Total Rated Power Output In kW (2001 - 2010)  \
count      2810.000000
mean      180.258767
std       300.729756
min         0.000000
25%         9.690000
50%        61.752000
75%       222.033500
max      3164.003000

      Jan 2011 - Rated Power Output In kW  \
count      2810.000000
mean      20.679269
std       42.076633
min         0.000000
25%         0.000000
50%         4.560000
75%       23.756000
max      754.484000

      Feb 2011 - Rated Power Output In kW  \
count      2810.000000
mean      22.982555
std       43.400588
min         0.000000
25%         0.000000
50%         6.110000
75%       26.801250
max      572.100000

      Mar 2011 - Rated Power Output In kW  \

```

count	2810.000000
mean	28.511664
std	51.380393
min	0.000000
25%	0.000000
50%	8.150000
75%	34.588750
max	566.300000

Apr 2011 - Rated Power Output In kW \

count	2810.000000
mean	27.042775
std	48.560529
min	0.000000
25%	0.000000
50%	7.512000
75%	32.378750
max	470.395000

May 2011 - Rated Power Output In kW \

count	2810.000000
mean	41.266189
std	75.581136
min	0.000000
25%	0.000000
50%	12.685000
75%	47.706000
max	770.705000

Jun 2011 - Rated Power Output In kW \

count	2810.000000
mean	59.553616
std	105.559607
min	0.000000
25%	2.752500
50%	19.337500
75%	70.321250
max	1225.978000

Jul 2011 - Rated Power Output In kW \

count	2810.000000
mean	19.254452
std	37.929121
min	0.000000
25%	0.000000
50%	3.995000
75%	20.558750

max	355.611000	
Aug 2011 - Rated Power Output In kW ... \		
count	2810.000000	...
mean	20.140893	...
std	39.546335	...
min	0.000000	...
25%	0.000000	...
50%	4.940000	...
75%	20.415000	...
max	516.160000	...
Nov 2024 - Rated Power Output In kW \		
count	2810.000000	
mean	96.599781	
std	171.460774	
min	0.000000	
25%	0.000000	
50%	32.552500	
75%	122.855000	
max	2179.230000	
Dec 2024 - Rated Power Output In kW \		
count	2810.000000	
mean	113.245737	
std	209.621779	
min	0.000000	
25%	0.000000	
50%	37.840000	
75%	139.858750	
max	3111.890000	
Jan 2025 - Rated Power Output In kW \		
count	2810.000000	
mean	63.394109	
std	124.060835	
min	0.000000	
25%	0.000000	
50%	18.500000	
75%	75.732500	
max	1861.760000	
Feb 2025 - Rated Power Output In kW \		
count	2810.000000	
mean	80.743511	
std	150.836322	
min	0.000000	

25%	0.000000
50%	25.960000
75%	96.346250
max	1974.510000

Mar 2025 - Rated Power Output In kW \	
count	2810.000000
mean	83.354635
std	149.111981
min	0.000000
25%	0.000000
50%	29.480000
75%	105.112500
max	2077.895000

Apr 2025 - Rated Power Output In kW \	
count	2810.000000
mean	70.931454
std	132.374325
min	0.000000
25%	0.000000
50%	22.575000
75%	85.360000
max	1790.775000

May 2025 - Rated Power Output In kW \	
count	2810.000000
mean	70.325591
std	132.808058
min	0.000000
25%	0.000000
50%	22.285000
75%	83.186250
max	1721.985000

Jun 2025 - Rated Power Output In kW \	
count	2810.000000
mean	52.031738
std	103.070242
min	0.000000
25%	0.000000
50%	13.200000
75%	57.765000
max	1287.500000

Jul 2025 - Rated Power Output In kW		Total Rated Power Output In kW
count	2810.000000	2810.000000

mean	42.526434	9575.233955
std	84.652508	15602.295013
min	0.000000	0.080000
25%	0.000000	769.946500
50%	10.450000	3572.203500
75%	48.681250	11895.593500
max	960.645000	165866.824000

[8 rows x 178 columns]

1.2 Data Cleaning and Preparation

Several preprocessing steps were performed before analysis:

- Unnecessary historical columns from 2001–2010 were removed,
- Postcode columns were renamed for clarity,
- monthly columns were reshaped from wide format into long format,
- postcodes were matched to Australian states and territories,
- invalid postcode entries were removed,
- dates were converted into yearly values for trend analysis.

2 Check for Missing Values

```
[3]: # Checking for missing values
missing_values_count = solar_capacity_df.isnull().sum()

# checking for percentage missing
total_cells = solar_capacity_df.size # rows x columns
total_missing = missing_values_count.sum()
percent_missing = (total_missing) * 100
print(percent_missing)
```

0

3 State / Territory

```
[16]: import pandas as pd

postcode_table = pd.DataFrame({
    "State/Territory": ["NSW", "ACT", "VIC", "QLD", "SA", "WA", "TAS", "NT"],
    "Postcode Range": [
        "2000-2599",
        "2600-2999",
        "3000-3999",
        "4000-4999",
        "5000-5999",
        "6000-6999",
    ]
})
```

```

        "7000-7999",
        "0800-0999",
    ]
})
postcode_table

```

```

[16]: State/Territory Postcode Range
0      NSW      2000-2599
1      ACT      2600-2999
2      VIC      3000-3999
3      QLD      4000-4999
4      SA       5000-5999
5      WA       6000-6999
6      TAS      7000-7999
7      NT       0800-0999

```

```

[17]: # Split the range into numeric start/end using postcode table
postcode_table[["start", "end"]] = (
    postcode_table["Postcode Range"].str.split("-", expand=True).astype(int)
)

# Save table as a csv file
postcode_table.to_csv("postcode_mapping.csv", index=False)
postcode_table.head()

```

```

[17]: State/Territory Postcode Range start end
0      NSW      2000-2599    2000  2599
1      ACT      2600-2999    2600  2999
2      VIC      3000-3999    3000  3999
3      QLD      4000-4999    4000  4999
4      SA       5000-5999    5000  5999

```

```

[18]: # Save table as a csv file
postcode_table.to_csv("postcode_mapping.csv", index=False)

```

3.1 Clean Columns for Solar Capacity (2011-2025)

```

[216]: # Strip column name for Dates strip from ( - Rated Power Output in kW)
solar_capacity_df.columns = (solar_capacity_df.columns.str.split("-").str[0])

# Rename first column to Postcode
solar_capacity_df = solar_capacity_df.rename(columns={"Small Unit Postcode": "Postcode"})

# Remove not used columns
# Historic Total Rated Power Output In kW (2001 - 2010)
solar_capacity_df = solar_capacity_df.drop(

```

```

        columns=[col for col in solar_capacity_df.columns if "2001" in col]
    )

    # Reshape the data format from wide -> long
    solar_long = solar_capacity_df.melt(
        id_vars=["Postcode"],
        var_name="date",
        value_name="capacity" )

    print(solar_long.head(10))

```

	Postcode	date	capacity
0	0	Jan 2011	0.0
1	200	Jan 2011	0.0
2	800	Jan 2011	0.0
3	801	Jan 2011	0.0
4	803	Jan 2011	0.0
5	804	Jan 2011	0.0
6	810	Jan 2011	3.02
7	811	Jan 2011	0.0
8	812	Jan 2011	3.0
9	813	Jan 2011	0.0

3.2 Merge Solar Capacity Data with their respective State/Territory Postcode

Combine solar capacity data trends with postcode-based State/Territory

```

[217]: # Check which Postcode it falls into

def get_state(pc):
    row = postcode_table[(pc >= postcode_table["start"]) & (pc <=
    ↪postcode_table["end"])]
    if not row.empty:
        return row["State/Territory"].iloc[0]
    return None

```

```

[218]: # Apply function to Solar Capacity dataframe
solar_capacity_df["State/Territory"] = (
    solar_capacity_df["Postcode"].astype(int).apply(get_state)
)
solar_capacity_df.head()

# Reshape the data format from wide -> long
solar_long = solar_capacity_df.melt(
    id_vars=["Postcode", "State/Territory"],
    var_name="date",
    value_name="capacity" )

```

```
print(solar_long.head())
```

	Postcode	State/Territory	date	capacity
0	0	NaN	Jan 2011	0.0
1	200	NaN	Jan 2011	0.0
2	800	NT	Jan 2011	0.0
3	801	NT	Jan 2011	0.0
4	803	NT	Jan 2011	0.0

```
[181]: # Drop Nan values for State/Territory -> drop postcode 0, 200
solar_long = solar_long.dropna(subset=["State/Territory"])
solar_long.head()
```

```
[181]:
```

	Postcode	State/Territory	date	capacity
2	800	NT	Jan 2011	0.00
3	801	NT	Jan 2011	0.00
4	803	NT	Jan 2011	0.00
5	804	NT	Jan 2011	0.00
6	810	NT	Jan 2011	3.02

4 Format Dates from Column to year

```
[220]: # Convert date to real datetime objects
solar_long["date"] = pd.to_datetime(solar_long["date"], errors="coerce")
solar_long["date"].isna().sum()
```

C:\Users\kabyw\AppData\Local\Temp\ipykernel_38488\858281054.py:2: UserWarning: Could not infer format, so each element will be parsed individually, falling back to `dateutil`. To ensure parsing is consistent and as-expected, please specify a format.

```
solar_long["date"] = pd.to_datetime(solar_long["date"], errors="coerce")
```

```
[220]: np.int64(2810)
```

```
[221]: # Extract yearly data
solar_long["year"] = solar_long["date"].dt.year

# Aggregate to yearly
solar_yearly = solar_long.groupby("year")["capacity"].sum().reset_index()
solar_yearly.head()

# Check to handle zero in dataset
# (solar_long["capacity"] == 0).mean() * 100 # gives np.float64(32.
# 04565674538984)
# solar_yearly = solar_yearly[solar_yearly["capacity"] > 0]
# print(solar_yearly)
```

```
[221]:      year      capacity
0  2011.0    872239.836
1  2012.0   1035750.287
2  2013.0    792188.984
3  2014.0    800261.301
4  2015.0    705967.095
```

5 Solar Capacity Trends by State/Territory

```
[253]: solar_long["State/Territory"]
solar_long.columns
```

```
[253]: Index(['Postcode', 'State/Territory', 'date', 'capacity', 'year', 'Year'],
      dtype='str')
```

```
[185]: solar_by_state = solar_long.groupby("State/Territory")["capacity"].sum().
      ↪reset_index()
```

6 Solar Capacity per State/Territory per year

```
[222]: solar_long["Year"] = solar_long["date"].dt.year

# Aggregate
solar_by_state_year = (
    solar_long.groupby(["State/Territory", "Year"])["capacity"].sum().
    ↪reset_index()
)
solar_by_state_year
```

```
[222]:      State/Territory      Year      capacity
0              ACT  2011.0    67875.753
1              ACT  2012.0    41189.284
2              ACT  2013.0    52857.500
3              ACT  2014.0    52175.184
4              ACT  2015.0    58740.550
..          ...      ...      ...
115           WA  2021.0   353317.798
116           WA  2022.0   286828.657
117           WA  2023.0   273494.216
118           WA  2024.0   285067.544
119           WA  2025.0   132369.570
```

```
[120 rows x 3 columns]
```

7 Solar Installation Data Trends

```
[187]: # Load CSV
solar_installation_df = pd.
    ↪read_csv('sgu-solar-installations-2011-to-present-and-totals.csv')
print(f"Loaded {len(solar_installation_df):,} rows ×
    ↪{len(solar_installation_df.columns)} columns")
solar_installation_df.head()
```

Loaded 2,810 rows × 178 columns

```
[187]: Small Unit Installation Postcode \
0      0
1     200
2     800
3     801
4     803

Historic Total Installation Quantity (2001 - 2010) \
0      4
1      1
2      7
3      2
4      0

Jan 2011 - Installation Quantity  Feb 2011 - Installation Quantity \
0      0      0
1      0      0
2      0      0
3      0      0
4      0      0

Mar 2011 - Installation Quantity  Apr 2011 - Installation Quantity \
0      0      0
1      0      0
2      0      0
3      0      0
4      0      0

May 2011 - Installation Quantity  Jun 2011 - Installation Quantity \
0      0      0
1      0      0
2      1      0
3      1      0
4      0      0

Jul 2011 - Installation Quantity  Aug 2011 - Installation Quantity ... \
0      0      0      ...
```

1	0	0	...
2	1	0	...
3	0	0	...
4	0	0	...

	Nov 2024 - Installation Quantity	Dec 2024 - Installation Quantity	\
0	0	0	
1	0	0	
2	2	0	
3	0	0	
4	0	0	

	Jan 2025 - Installation Quantity	Feb 2025 - Installation Quantity	\
0	0	0	
1	0	0	
2	0	0	
3	0	0	
4	0	0	

	Mar 2025 - Installation Quantity	Apr 2025 - Installation Quantity	\
0	0	0	
1	0	0	
2	3	1	
3	0	0	
4	0	0	

	May 2025 - Installation Quantity	Jun 2025 - Installation Quantity	\
0	0	0	
1	0	0	
2	0	2	
3	0	0	
4	0	0	

	Jul 2025 - Installation Quantity	Total Installation Quantity
0	0	4
1	0	1
2	0	147
3	0	12
4	0	2

[5 rows x 178 columns]

```
[188]: solar_installation_df.info()
solar_installation_df.describe()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 2810 entries, 0 to 2809
Columns: 178 entries, Small Unit Installation Postcode to Total Installation
```

Quantity
dtypes: int64(178)
memory usage: 3.8 MB

```
[188]:      Small Unit Installation Postcode  \
count      2810.000000
mean      4168.266548
std      1592.968085
min         0.000000
25%      2845.250000
50%      3929.500000
75%      5340.750000
max      9729.000000

      Historic Total Installation Quantity (2001 - 2010)  \
count      2810.000000
mean      100.822420
std      173.362941
min         0.000000
25%         4.250000
50%        32.000000
75%       123.000000
max      2035.000000

      Jan 2011 - Installation Quantity  Feb 2011 - Installation Quantity  \
count      2810.000000      2810.000000
mean         9.062633      10.266904
std      19.484976      20.600454
min         0.000000      0.000000
25%         0.000000      0.000000
50%         2.000000      2.500000
75%        10.000000      12.000000
max      442.000000      309.000000

      Mar 2011 - Installation Quantity  Apr 2011 - Installation Quantity  \
count      2810.000000      2810.000000
mean      12.643416      11.807473
std      24.161736      22.295651
min         0.000000      0.000000
25%         0.000000      0.000000
50%         3.000000      3.000000
75%        14.750000      13.000000
max      273.000000      244.000000

      May 2011 - Installation Quantity  Jun 2011 - Installation Quantity  \
count      2810.000000      2810.000000
mean      17.896441      24.507829
```


std	34.868082	44.076386
min	0.000000	0.000000
25%	0.000000	1.000000
50%	5.000000	7.000000
75%	19.000000	28.000000
max	451.000000	462.000000

	Jul 2011 - Installation Quantity	Aug 2011 - Installation Quantity \
count	2810.000000	2810.000000
mean	7.907473	7.874377
std	16.140359	15.743275
min	0.000000	0.000000
25%	0.000000	0.000000
50%	2.000000	2.000000
75%	8.000000	8.000000
max	156.000000	207.000000

	... Nov 2024 - Installation Quantity \
count	2810.000000
mean	9.459431
std	17.113820
min	0.000000
25%	0.000000
50%	3.000000
75%	12.000000
max	199.000000

	Dec 2024 - Installation Quantity	Jan 2025 - Installation Quantity \
count	2810.000000	2810.000000
mean	9.884698	6.762989
std	18.781449	13.517234
min	0.000000	0.000000
25%	0.000000	0.000000
50%	3.000000	2.000000
75%	12.000000	8.000000
max	313.000000	225.000000

	Feb 2025 - Installation Quantity	Mar 2025 - Installation Quantity \
count	2810.000000	2810.000000
mean	8.264057	8.216726
std	15.683298	15.280305
min	0.000000	0.000000
25%	0.000000	0.000000
50%	3.000000	3.000000
75%	10.000000	10.000000
max	218.000000	226.000000

	Apr 2025 - Installation Quantity	May 2025 - Installation Quantity \
count	2810.000000	2810.000000
mean	7.175445	7.114235
std	13.759886	14.048666
min	0.000000	0.000000
25%	0.000000	0.000000
50%	2.000000	2.000000
75%	8.000000	8.000000
max	214.000000	227.000000

	Jun 2025 - Installation Quantity	Jul 2025 - Installation Quantity \
count	2810.000000	2810.000000
mean	5.438790	4.364413
std	11.169237	8.858543
min	0.000000	0.000000
25%	0.000000	0.000000
50%	1.000000	1.000000
75%	6.000000	5.000000
max	160.000000	112.000000

	Total Installation Quantity
count	2810.000000
mean	1485.610676
std	2484.238328
min	1.000000
25%	104.250000
50%	522.500000
75%	1836.750000
max	24716.000000

[8 rows x 178 columns]

```
[189]: # Checking for missing values
missing_values_count = solar_installation_df.isnull().sum()

# checking for percentage missing
total_cells = solar_installation_df.size # rows x columns
total_missing = missing_values_count.sum()
percent_missing = (total_missing) * 100
print(percent_missing)
```

0

7.1 Format Data for Solar Installation (2011-2025)

```
[192]: # Strip column name for Dates strip from ( - Rated Power Output in kW)
solar_installation_df.columns = (solar_installation_df.columns.str.split("-").
    ↪str[0])

# Rename first column to Postcode
solar_installation_df = solar_installation_df.rename(columns={"Small Unit_↪
    ↪Installation Postcode": "Postcode"})

# Remove not used columns
# Historic Total Rated Power Output In kW (2001 - 2010)
solar_installation_df = solar_installation_df.drop(
    columns=[col for col in solar_installation_df.columns if "2001" in col]
)

# Reshape the data format from wide -> long
solar_installation_long = solar_installation_df.melt(
    id_vars=["Postcode"],
    var_name="date",
    value_name="quantity" )

print(solar_installation_long.head(10))
```

	Postcode	date	quantity
0	0	Jan 2011	0
1	200	Jan 2011	0
2	800	Jan 2011	0
3	801	Jan 2011	0
4	803	Jan 2011	0
5	804	Jan 2011	0
6	810	Jan 2011	2
7	811	Jan 2011	0
8	812	Jan 2011	1
9	813	Jan 2011	0

7.2 Merge Solar Installation Data with their respective State/Territory Postcode

Combine solar installation data trends with postcode-based State/Territory

```
[194]: # Apply function to Solar Installation dataframe
solar_installation_df["State/Territory"] = (
    solar_installation_df["Postcode"].astype(int).apply(get_state)
)
solar_installation_df.head()

# Reshape the data format from wide -> long
```

```
solar_installation_long = solar_installation_df.melt(
    id_vars=["Postcode", "State/Territory"],
    var_name="date",
    value_name="quantity" )
```

```
print(solar_installation_long.head())
```

	Postcode	State/Territory	date	quantity
0	0	NaN	Jan 2011	0
1	200	NaN	Jan 2011	0
2	800	NT	Jan 2011	0
3	801	NT	Jan 2011	0
4	803	NT	Jan 2011	0

```
[195]: # Drop Nan values for State/Territory -> drop postcode 0, 200
solar_installation_long = solar_installation_long.dropna(subset=["State/
↳ Territory"])
solar_installation_long.head()
```

```
[195]:
```

	Postcode	State/Territory	date	quantity
2	800	NT	Jan 2011	0
3	801	NT	Jan 2011	0
4	803	NT	Jan 2011	0
5	804	NT	Jan 2011	0
6	810	NT	Jan 2011	2

7.2.1 Clean Dates Column to year

```
[196]: # Convert date to real datetime objects
solar_installation_long["date"] = pd.
↳ to_datetime(solar_installation_long["date"], errors="coerce")
solar_installation_long["date"].isna().sum()

# Extract yearly data
solar_installation_long["year"] = solar_installation_long["date"].dt.year

# Aggregate to yearly
solar_yearly = solar_installation_long.groupby("year")["quantity"].sum().
↳ reset_index()
solar_yearly.head(25)
```

C:\Users\kabyw\AppData\Local\Temp\ipykernel_38488\1796411224.py:2: UserWarning:
Could not infer format, so each element will be parsed individually, falling
back to `dateutil`. To ensure parsing is consistent and as-expected, please
specify a format.

```
solar_installation_long["date"] =
pd.to_datetime(solar_installation_long["date"], errors="coerce")
```

```
[196]:
```

	year	quantity
0	2011.0	360730
1	2012.0	343302
2	2013.0	200406
3	2014.0	180136
4	2015.0	141494
5	2016.0	132692
6	2017.0	174939
7	2018.0	224842
8	2019.0	284028
9	2020.0	370315
10	2021.0	377457
11	2022.0	315715
12	2023.0	333826
13	2024.0	318249
14	2025.0	133014

7.3 Solar Installation Trends by State/Territory

```
[197]: solar_installation_long["State/Territory"]
solar_installation_long.columns
```

```
[197]: Index(['Postcode', 'State/Territory', 'date', 'quantity', 'year'], dtype='str')
```

```
[198]: solar_installation_by_state = solar_installation_long.groupby("State/
↳ Territory")["quantity"].sum().reset_index()
```

7.4 Solar Installation per State/Territory per Year

This data shows how solar installation numbers have evolved around states from 2011 to 2025.

```
[199]: solar_installation_long["Year"] = solar_long["date"].dt.year

# Aggregate
solar_installation_by_state_year = (
    solar_installation_long.groupby(["State/Territory", "Year"])["quantity"].
    ↳sum().reset_index()
)
solar_installation_by_state_year
```

```
[199]:
```

	State/Territory	Year	quantity
0	ACT	2011.0	24774
1	ACT	2012.0	13426
2	ACT	2013.0	12556
3	ACT	2014.0	10783
4	ACT	2015.0	10277
..	...	...	...
115	WA	2021.0	48780

116	WA	2022.0	38691
117	WA	2023.0	35213
118	WA	2024.0	35535
119	WA	2025.0	16450

[120 rows x 3 columns]

8 Australia's Populations table from ABS - All states

```
[14]: %pip install openpyxl
```

Collecting openpyxl

Downloading openpyxl-3.1.5-py2.py3-none-any.whl.metadata (2.5 kB)

Collecting et-xmlfile (from openpyxl)

Downloading et_xmlfile-2.0.0-py3-none-any.whl.metadata (2.7 kB)

Downloading openpyxl-3.1.5-py2.py3-none-any.whl (250 kB)

Downloading et_xmlfile-2.0.0-py3-none-any.whl (18 kB)

Installing collected packages: et-xmlfile, openpyxl

```

----- 1/2 [openpyxl]
----- 1/2 [openpyxl]
----- 1/2 [openpyxl]
----- 1/2 [openpyxl]
----- 1/2 [openpyxl]
----- 1/2 [openpyxl]
----- 1/2 [openpyxl]
----- 1/2 [openpyxl]
----- 1/2 [openpyxl]
----- 1/2 [openpyxl]
----- 2/2 [openpyxl]

```

Successfully installed et-xmlfile-2.0.0 openpyxl-3.1.5

Note: you may need to restart the kernel to use updated packages.

```
[73]: population_data = pd.DataFrame(pd.read_excel("Australia_population.xlsx"))
      population_data.head()
```

```
[73]:      Unnamed: 0  \
0      Unit
1      Series Type
2      Data Type
3      Frequency
4      Collection Month
```

```

Estimated Resident Population ; Male ; New South Wales ;  \
0      Persons
1      Original
2      STOCK_CLOSE

```

3	Quarter
4	3

Estimated Resident Population ; Male ; Victoria ; \
0 Persons
1 Original
2 STOCK_CLOSE
3 Quarter
4 3

Estimated Resident Population ; Male ; Queensland ; \
0 Persons
1 Original
2 STOCK_CLOSE
3 Quarter
4 3

Estimated Resident Population ; Male ; South Australia ; \
0 Persons
1 Original
2 STOCK_CLOSE
3 Quarter
4 3

Estimated Resident Population ; Male ; Western Australia ; \
0 Persons
1 Original
2 STOCK_CLOSE
3 Quarter
4 3

Estimated Resident Population ; Male ; Tasmania ; \
0 Persons
1 Original
2 STOCK_CLOSE
3 Quarter
4 3

Estimated Resident Population ; Male ; Northern Territory ; \
0 Persons
1 Original
2 STOCK_CLOSE
3 Quarter
4 3

Estimated Resident Population ; Male ; Australian Capital Territory ; \
0 Persons

1	Original	
2	STOCK_CLOSE	
3	Quarter	
4	3	

Estimated Resident Population ; Male ; Australia ; ... \		
0	Persons	...
1	Original	...
2	STOCK_CLOSE	...
3	Quarter	...
4	3	...

Estimated Resident Population ; Female ; Australia ; \		
0	Persons	
1	Original	
2	STOCK_CLOSE	
3	Quarter	
4	3	

Estimated Resident Population ; Persons ; New South Wales ; \		
0	Persons	
1	Original	
2	STOCK_CLOSE	
3	Quarter	
4	3	

Estimated Resident Population ; Persons ; Victoria ; \		
0	Persons	
1	Original	
2	STOCK_CLOSE	
3	Quarter	
4	3	

Estimated Resident Population ; Persons ; Queensland ; \		
0	Persons	
1	Original	
2	STOCK_CLOSE	
3	Quarter	
4	3	

Estimated Resident Population ; Persons ; South Australia ; \		
0	Persons	
1	Original	
2	STOCK_CLOSE	
3	Quarter	
4	3	


```

    Estimated Resident Population ; Persons ; Western Australia ; \
0                                     Persons
1                                     Original
2                                     STOCK_CLOSE
3                                     Quarter
4                                     3

```

```

    Estimated Resident Population ; Persons ; Tasmania ; \
0                                     Persons
1                                     Original
2                                     STOCK_CLOSE
3                                     Quarter
4                                     3

```

```

    Estimated Resident Population ; Persons ; Northern Territory ; \
0                                     Persons
1                                     Original
2                                     STOCK_CLOSE
3                                     Quarter
4                                     3

```

```

    Estimated Resident Population ; Persons ; Australian Capital Territory ; \
0                                     Persons
1                                     Original
2                                     STOCK_CLOSE
3                                     Quarter
4                                     3

```

```

    Estimated Resident Population ; Persons ; Australia ;
0                                     Persons
1                                     Original
2                                     STOCK_CLOSE
3                                     Quarter
4                                     3

```

```
[5 rows x 28 columns]
```

```
[74]: # check dataset
population_data.info()
population_data.describe()
```

```

<class 'pandas.DataFrame'>
RangeIndex: 187 entries, 0 to 186
Data columns (total 28 columns):
#   Column
Non-Null Count  Dtype
---  -
-----

```

```

0   Unnamed: 0
187 non-null    object
1   Estimated Resident Population ; Male ; New South Wales ;
187 non-null    object
2   Estimated Resident Population ; Male ; Victoria ;
187 non-null    object
3   Estimated Resident Population ; Male ; Queensland ;
187 non-null    object
4   Estimated Resident Population ; Male ; South Australia ;
187 non-null    object
5   Estimated Resident Population ; Male ; Western Australia ;
187 non-null    object
6   Estimated Resident Population ; Male ; Tasmania ;
187 non-null    object
7   Estimated Resident Population ; Male ; Northern Territory ;
187 non-null    object
8   Estimated Resident Population ; Male ; Australian Capital Territory ;
187 non-null    object
9   Estimated Resident Population ; Male ; Australia ;
187 non-null    object
10  Estimated Resident Population ; Female ; New South Wales ;
187 non-null    object
11  Estimated Resident Population ; Female ; Victoria ;
187 non-null    object
12  Estimated Resident Population ; Female ; Queensland ;
187 non-null    object
13  Estimated Resident Population ; Female ; South Australia ;
187 non-null    object
14  Estimated Resident Population ; Female ; Western Australia ;
187 non-null    object
15  Estimated Resident Population ; Female ; Tasmania ;
187 non-null    object
16  Estimated Resident Population ; Female ; Northern Territory ;
187 non-null    object
17  Estimated Resident Population ; Female ; Australian Capital Territory ;
187 non-null    object
18  Estimated Resident Population ; Female ; Australia ;
187 non-null    object
19  Estimated Resident Population ; Persons ; New South Wales ;
187 non-null    object
20  Estimated Resident Population ; Persons ; Victoria ;
187 non-null    object
21  Estimated Resident Population ; Persons ; Queensland ;
187 non-null    object
22  Estimated Resident Population ; Persons ; South Australia ;
187 non-null    object
23  Estimated Resident Population ; Persons ; Western Australia ;
187 non-null    object

```

```

24 Estimated Resident Population ; Persons ; Tasmania ;
187 non-null      object
25 Estimated Resident Population ; Persons ; Northern Territory ;
187 non-null      object
26 Estimated Resident Population ; Persons ; Australian Capital Territory ;
187 non-null      object
27 Estimated Resident Population ; Persons ; Australia ;
187 non-null      object
dtypes: object(28)
memory usage: 41.0+ KB

```

```

[74]:      Unnamed: 0 Estimated Resident Population ; Male ; New South Wales ; \
count          187                                187
unique          187                                187
top            Unit                                Persons
freq           1                                  1

```

```

      Estimated Resident Population ; Male ; Victoria ; \
count          187
unique          187
top            Persons
freq           1

```

```

      Estimated Resident Population ; Male ; Queensland ; \
count          187
unique          187
top            Persons
freq           1

```

```

      Estimated Resident Population ; Male ; South Australia ; \
count          187
unique          187
top            Persons
freq           1

```

```

      Estimated Resident Population ; Male ; Western Australia ; \
count          187
unique          187
top            Persons
freq           1

```

```

      Estimated Resident Population ; Male ; Tasmania ; \
count          187
unique          187
top            Persons
freq           1

```

	Estimated Resident Population ; Male ; Northern Territory ; \	
count	187	
unique	187	
top	Persons	
freq	1	

	Estimated Resident Population ; Male ; Australian Capital Territory ; \	
count	187	
unique	187	
top	Persons	
freq	1	

	Estimated Resident Population ; Male ; Australia ; ... \	
count	187	...
unique	187	...
top	Persons	...
freq	1	...

	Estimated Resident Population ; Female ; Australia ; \	
count	187	
unique	187	
top	Persons	
freq	1	

	Estimated Resident Population ; Persons ; New South Wales ; \	
count	187	
unique	187	
top	Persons	
freq	1	

	Estimated Resident Population ; Persons ; Victoria ; \	
count	187	
unique	187	
top	Persons	
freq	1	

	Estimated Resident Population ; Persons ; Queensland ; \	
count	187	
unique	187	
top	Persons	
freq	1	

	Estimated Resident Population ; Persons ; South Australia ; \	
count	187	
unique	187	
top	Persons	

```

freq                                                    1

    Estimated Resident Population ; Persons ; Western Australia ; \
count                                                    187
unique                                                    187
top                                                       Persons
freq                                                    1

    Estimated Resident Population ; Persons ; Tasmania ; \
count                                                    187
unique                                                    186
top                                                       473430
freq                                                    2

    Estimated Resident Population ; Persons ; Northern Territory ; \
count                                                    187
unique                                                    187
top                                                       Persons
freq                                                    1

    Estimated Resident Population ; Persons ; Australian Capital Territory
; \
count                                                    187
unique                                                    187
top                                                       Persons
freq                                                    1

    Estimated Resident Population ; Persons ; Australia ;
count                                                    187
unique                                                    187
top                                                       Persons
freq                                                    1

[4 rows x 28 columns]

```

9 Remove Unwanted data from Columns

```

[110]: # Clean data and drop metadata
population_data["Date"] = pd.to_datetime(
    population_data["Unnamed: 0"],
    format="%b-%Y",
    errors="coerce"
)
population_data = population_data.dropna(subset=["Date"])
# population_data

```

```
[113]: # filter year from 2011-2025
population_data = population_data[
    (population_data["Date"].dt.year >= 2011) &
    (population_data["Date"].dt.year <= 2025)
]

# keep 6 months rows so June
population_yearly = population_data[population_data["Date"].dt.month == 6]

# Extract only "Persons" columns
persons_cols = [col for col in population_yearly.columns if col.
    ↪startswith("Persons")]
population_persons = population_yearly[persons_cols].copy()
population_persons
```

```
[113]: Persons ; New South Wales ; Persons ; Victoria ; \
129          7218529          5537817
133          7304244          5651091
137          7404032          5772669
141          7508353          5894917
145          7616168          6022322
149          7732858          6173172
153          7855316          6302608
157          7954476          6423038
161          8046748          6537305
165          8110610          6615046
169          8097062          6547822
173          8182098          6615232
177          8356387          6797169
181          8492050          6950961
185          8592524          7073859

Persons ; Queensland ; Persons ; South Australia ; \
129          4476778          1639614
133          4568687          1656725
137          4652824          1671488
141          4719653          1686945
145          4777692          1700668
149          4845152          1712843
153          4926380          1728673
157          5006623          1746137
161          5088847          1767395
165          5165613          1790355
169          5215814          1802601
173          5310905          1824969
177          5450235          1857646
181          5571890          1882164
```

185	5669764	1902665
-----	---------	---------

	Persons ;	Western Australia ;	Persons ;	Tasmania ;	\
129		2353409		511483	
133		2425507		511724	
137		2486944		512231	
141		2517608		513621	
145		2540672		515117	
149		2555978		517514	
153		2585720		526762	
157		2617792		537291	
161		2659625		547841	
165		2712912		557578	
169		2749365		567239	
173		2793934		572300	
177		2890936		573731	
181		2978147		574765	
185		3044382		576261	

	Persons ;	Northern Territory ;	Persons ;	Australian Capital Territory ;	\
129		231292		367985	
133		235915		376539	
137		241722		383257	
141		242894		388799	
145		244692		395813	
149		245678		403104	
153		247412		415046	
157		247095		426081	
161		246559		435730	
165		247428		444903	
169		248151		452508	
173		253100		461262	
177		257508		471340	
181		260884		478430	
185		264556		484630	

	Persons ;	Australia ;
129		22340024
133		22733465
137		23128129
141		23475686
145		23815995
149		24190907
153		24592588
157		24963258
161		25334826
165		25649248

```

169                25685412
173                26018721
177                26659922
181                27194286
185                27613654

```

```

[137]: # Subset the DataFrame to only "Persons" column
cleaned_cols = {}
for col in persons_cols:
    parts = col.split(";") # split column name
    parts = [p.strip() for p in parts if p.strip()] # remove empty spaces
    state_name = parts[-1] # last part is state/territory
    cleaned_cols[col] = state_name # original -> cleaned

# rename the column
population_persons = population_persons.rename(columns=cleaned_cols)

# population_persons

```

10 Merge clean Population Dataset

```

[138]: # Add Year column
population_yearly["Year"] = population_yearly["Date"].dt.year
population_persons["Year"] = population_yearly["Year"].values

population_merged = population_yearly[["Date", "Year"]].merge(
    population_persons,
    on="Year",
    how="left"
)
population_merged

```

```

[138]:      Date  Year New South Wales Victoria Queensland South Australia \
0  2011-06-01  2011      7218529  5537817    4476778      1639614
1  2012-06-01  2012      7304244  5651091    4568687      1656725
2  2013-06-01  2013      7404032  5772669    4652824      1671488
3  2014-06-01  2014      7508353  5894917    4719653      1686945
4  2015-06-01  2015      7616168  6022322    4777692      1700668
5  2016-06-01  2016      7732858  6173172    4845152      1712843
6  2017-06-01  2017      7855316  6302608    4926380      1728673
7  2018-06-01  2018      7954476  6423038    5006623      1746137
8  2019-06-01  2019      8046748  6537305    5088847      1767395
9  2020-06-01  2020      8110610  6615046    5165613      1790355
10 2021-06-01  2021      8097062  6547822    5215814      1802601
11 2022-06-01  2022      8182098  6615232    5310905      1824969
12 2023-06-01  2023      8356387  6797169    5450235      1857646

```


13	2024-06-01	2024	8492050	6950961	5571890	1882164
14	2025-06-01	2025	8592524	7073859	5669764	1902665

	Western Australia	Tasmania	Northern Territory	Australian Capital Territory	\
0	2353409	511483	231292		367985
1	2425507	511724	235915		376539
2	2486944	512231	241722		383257
3	2517608	513621	242894		388799
4	2540672	515117	244692		395813
5	2555978	517514	245678		403104
6	2585720	526762	247412		415046
7	2617792	537291	247095		426081
8	2659625	547841	246559		435730
9	2712912	557578	247428		444903
10	2749365	567239	248151		452508
11	2793934	572300	253100		461262
12	2890936	573731	257508		471340
13	2978147	574765	260884		478430
14	3044382	576261	264556		484630

	Australia
0	22340024
1	22733465
2	23128129
3	23475686
4	23815995
5	24190907
6	24592588
7	24963258
8	25334826
9	25649248
10	25685412
11	26018721
12	26659922
13	27194286
14	27613654

11 Merge solar_by_state_year with population_merged

```
[232]: state_map = {
    "NSW": "New South Wales",
    "VIC": "Victoria",
    "QLD": "Queensland",
    "SA": "South Australia",
    "WA": "Western Australia",
    "TAS": "Tasmania",
```

```

    "NT": "Northern Territory",
    "ACT": "Australian Capital Territory"
}

```

```

[233]: solar_by_state_year["State_Full"] = solar_by_state_year["State/Territory"].
        ↪map(state_map)

```

12 Solar Capacity per Year with Population and State

To better understand solar energy adoption across Australia, solar capacity data was combined with state population estimates. This allowed the analysis to compare total installed solar capacity relative to population size rather than relying only on raw totals.

The population dataset was first transformed into long format to align with the yearly solar capacity records. The datasets were then merged using: * Year * State/Territory * Full state name

This produced a combined dataset containing * solar capacity, * annual population estimates

```

[234]: # convert to long format
population_capacity_long = population_merged.melt(
    id_vars=["Year", "Date"],
    var_name="State_Full",
    value_name="Population"
)

```

```

[235]: population_capacity_long["State_Full"].unique()
population_capacity_long

```

```

[235]:
   Year      Date      State_Full  Population
0  2011  2011-06-01  New South Wales    7218529
1  2012  2012-06-01  New South Wales    7304244
2  2013  2013-06-01  New South Wales    7404032
3  2014  2014-06-01  New South Wales    7508353
4  2015  2015-06-01  New South Wales    7616168
..   ...      ...
130 2021  2021-06-01      Australia    25685412
131 2022  2022-06-01      Australia    26018721
132 2023  2023-06-01      Australia    26659922
133 2024  2024-06-01      Australia    27194286
134 2025  2025-06-01      Australia    27613654

```

[135 rows x 4 columns]

```

[236]: # merge solar capacity + population
solar_with_capacity_population = solar_by_state_year.merge(
    population_capacity_long[["Year", "State_Full", "Population"]],
    on=["Year", "State_Full"],
    how="left"
)

```

```
)
solar_with_capacity_population
```

```
[236]:
```

	State/Territory	Year	capacity	State_Full \
0	ACT	2011.0	67875.753	Australian Capital Territory
1	ACT	2012.0	41189.284	Australian Capital Territory
2	ACT	2013.0	52857.500	Australian Capital Territory
3	ACT	2014.0	52175.184	Australian Capital Territory
4	ACT	2015.0	58740.550	Australian Capital Territory
..	...	...	...	...
115	WA	2021.0	353317.798	Western Australia
116	WA	2022.0	286828.657	Western Australia
117	WA	2023.0	273494.216	Western Australia
118	WA	2024.0	285067.544	Western Australia
119	WA	2025.0	132369.570	Western Australia

	Population
0	367985
1	376539
2	383257
3	388799
4	395813
..	...
115	2749365
116	2793934
117	2890936
118	2978147
119	3044382

[120 rows x 5 columns]

```
[ ]: import matplotlib.pyplot as plt

# Pick a year to visualize
year_to_plot = 2020

# Filter the dataset
subset = solar_combined[solar_combined["Year"] == year_to_plot]

# Sort for nicer plotting
subset = subset.sort_values("capacity", ascending=False)

plt.figure(figsize=(12, 6))
plt.bar(subset["State_Full"], subset["capacity"], color="steelblue")

plt.title(f"Total Solar Capacity by State in {year_to_plot}")
plt.xlabel("State/Territory")
```

```
plt.ylabel("Installed Capacity (kW)")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
```

13 Solar Installation per Year with Population and State

```
[237]: solar_installation_by_state_year["State_Full"] = solar_by_state_year["State/
↳ Territory"].map(state_map)
```

```
[238]: # convert to long format
population_installation_long = population_merged.melt(
    id_vars=["Year", "Date"],
    var_name="State_Full",
    value_name="Population"
)

population_installation_long["State_Full"].unique()
population_installation_long
```

```
[238]:
```

	Year	Date	State_Full	Population
0	2011	2011-06-01	New South Wales	7218529
1	2012	2012-06-01	New South Wales	7304244
2	2013	2013-06-01	New South Wales	7404032
3	2014	2014-06-01	New South Wales	7508353
4	2015	2015-06-01	New South Wales	7616168
..	...	...	...	...
130	2021	2021-06-01	Australia	25685412
131	2022	2022-06-01	Australia	26018721
132	2023	2023-06-01	Australia	26659922
133	2024	2024-06-01	Australia	27194286
134	2025	2025-06-01	Australia	27613654

[135 rows x 4 columns]

```
[239]: # merge solar installation + population
solar_with_quantity_population = solar_installation_by_state_year.merge(
    population_installation_long[["Year", "State_Full", "Population"]],
    on=["Year", "State_Full"],
    how="left"
)
solar_with_quantity_population
```

```
[239]:
```

	State/Territory	Year	quantity	State_Full	Population
0	ACT	2011.0	24774	Australian Capital Territory	367985
1	ACT	2012.0	13426	Australian Capital Territory	376539

2	ACT	2013.0	12556	Australian Capital Territory	383257
3	ACT	2014.0	10783	Australian Capital Territory	388799
4	ACT	2015.0	10277	Australian Capital Territory	395813
..	...	...	...	...	...
115	WA	2021.0	48780	Western Australia	2749365
116	WA	2022.0	38691	Western Australia	2793934
117	WA	2023.0	35213	Western Australia	2890936
118	WA	2024.0	35535	Western Australia	2978147
119	WA	2025.0	16450	Western Australia	3044382

[120 rows x 5 columns]

```
[240]: # Merge Solar capacity + solar installation + population / year Data
solar_installation_df.columns
```

```
[240]: Index(['Postcode', 'Jan 2011 ', 'Feb 2011 ', 'Mar 2011 ', 'Apr 2011 ',
        'May 2011 ', 'Jun 2011 ', 'Jul 2011 ', 'Aug 2011 ', 'Sep 2011 ',
        ...,
        'Dec 2024 ', 'Jan 2025 ', 'Feb 2025 ', 'Mar 2025 ', 'Apr 2025 ',
        'May 2025 ', 'Jun 2025 ', 'Jul 2025 ', 'Total Installation Quantity',
        'State/Territory'],
        dtype='str', length=178)
```

```
[242]: # merge capacity + installation
solar_combined = solar_with_capacity_population.merge(
    solar_with_quantity_population[["State_Full", "Year", "quantity"]],
    on=["State_Full", "Year"],
    how="outer" )

solar_combined
```

```
[242]: State/Territory  Year  capacity  State_Full \
0          ACT  2011.0  67875.753  Australian Capital Territory
1          ACT  2012.0  41189.284  Australian Capital Territory
2          ACT  2013.0  52857.500  Australian Capital Territory
3          ACT  2014.0  52175.184  Australian Capital Territory
4          ACT  2015.0  58740.550  Australian Capital Territory
..          ...      ...      ...      ...
115         WA  2021.0  353317.798  Western Australia
116         WA  2022.0  286828.657  Western Australia
117         WA  2023.0  273494.216  Western Australia
118         WA  2024.0  285067.544  Western Australia
119         WA  2025.0  132369.570  Western Australia

Population  quantity
0      367985      24774
1      376539      13426
```

2	383257	12556
3	388799	10783
4	395813	10277
..	...	...
115	2749365	48780
116	2793934	38691
117	2890936	35213
118	2978147	35535
119	3044382	16450

[120 rows x 6 columns]

13.0.1 Solar Capacity by State (Bar Chart)

This bar chart shows the total installed solar capacity (measured in kilowatts) for each Australian state and territory in a selected year.

```
[251]: import matplotlib.pyplot as plt

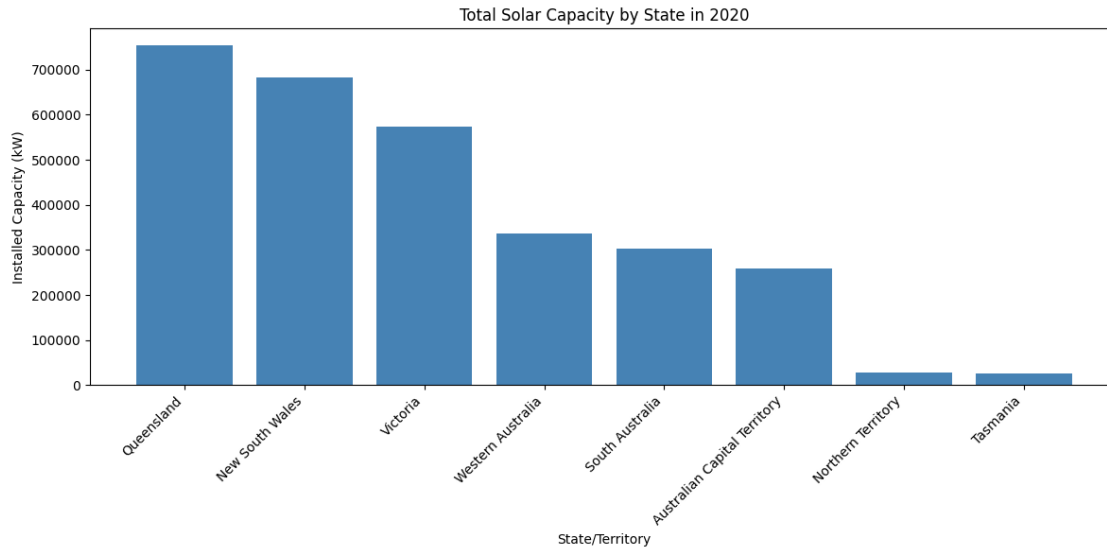
# Pick a year to visualize
year_to_plot = 2020

# Filter the dataset
subset = solar_combined[solar_combined["Year"] == year_to_plot]

# Sort for nicer plotting
subset = subset.sort_values("capacity", ascending=False)

plt.figure(figsize=(12, 6))
plt.bar(subset["State_Full"], subset["capacity"], color="steelblue")

plt.title(f"Total Solar Capacity by State in {year_to_plot}")
plt.xlabel("State/Territory")
plt.ylabel("Installed Capacity (kW)")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
```



13.0.2 Key Findings

Larger States Recorded the Highest Total Capacity

The chart shows that: * New South Wales, * Queensland, * Victoria, recorded the highest levels of installed solar capacity in 2020. These states have the largest populations in Australia, which contributes to their high total solar output.

Queensland Demonstrated Strong Solar Adoption

Queensland recorded particularly strong solar capacity levels, reflecting the state's favourable climate conditions and high levels of household solar participation.

Smaller Territories Recorded Lower Totals

The Australian Capital Territory and Northern Territory showed lower total installed capacity compared to larger states. This is largely due to their smaller populations and fewer residential areas.

National Growth in Renewable Energy

The overall distribution of solar capacity across states indicates that renewable energy adoption had become widespread by 2020. Nearly every state demonstrated substantial installed capacity, highlighting Australia's increasing reliance on solar energy technologies.

13.0.3 Solar Installations by State (Bar Chart)

This bar chart displays the number of solar system installations recorded in each Australian state and territory for a selected year.

```
[1]: import matplotlib.pyplot as plt
```

```

year_to_plot = 2020

subset = solar_combined[solar_combined["Year"] == year_to_plot]
subset = subset.sort_values("quantity", ascending=False)

plt.figure(figsize=(12, 6))
plt.bar(subset["State_Full"], subset["quantity"], color="orange")

plt.title(f"Solar Installations (Quantity) by State in {year_to_plot}")
plt.xlabel("State/Territory")
plt.ylabel("Number of Installations")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()

```

```

-----
NameError                                Traceback (most recent call last)
Cell In[1], line 5
      1 import matplotlib.pyplot as plt
      2
      3 year_to_plot = 2020
      4
----> 5 subset = solar_combined[solar_combined["Year"] == year_to_plot]
      6 subset = subset.sort_values("quantity", ascending=False)
      7
      8 plt.figure(figsize=(12, 6))

NameError: name 'solar_combined' is not defined

```

13.0.4 Key Findings

Queensland and New South Wales Led Installation Activity

Queensland and New South Wales recorded the highest numbers of solar installations during 2020. This suggests strong household participation in renewable energy adoption and increasing demand for residential solar systems.

Installation Growth Reflects Rising Demand for Renewable Energy

The high installation numbers observed across multiple states indicate that solar energy adoption accelerated significantly during this period. Several factors may have contributed to this growth, including:

- increasing electricity prices,
- government rebates and incentives,
- improved affordability of solar technology,
- growing public awareness of environmental sustainability.

Smaller Regions Had Lower Installation Numbers States and territories with smaller popu-

lations, such as Tasmania and the Northern Territory, recorded lower installation totals. However, lower totals do not necessarily indicate weaker adoption rates, as population size strongly affects raw installation counts.

14 Australia's Solar Capacity, Installations and Population by State (2011-2025)

This dataset combines three key sources of information to provide a comprehensive view of solar energy adoption across Australian states and territories from 2011 to 2025:

- **Solar Capacity (kW):** Total installed solar capacity recorded each year.
- **Solar Installations (Quantity):** Number of new solar systems installed annually.
- **Population:** State-level population estimates used to standardize solar metrics.

To enable fair comparison across states with different population sizes, the dataset includes population-standardized indicators:

- **Capacity per Person:** Installed solar capacity divided by population.
- **Installations per Person:** Number of installations divided by population.
- **Capacity per 100,000 People:** A normalized metric commonly used in energy and public reporting.
- **Installations per 100,000 People:** Standardized installation rate.

These metrics allow for meaningful cross-state comparisons and trend analysis, highlighting how solar adoption has evolved relative to population growth.

```
[244]: solar_final = solar_combined[
        ["State_Full", "State/Territory", "Year",
         "capacity", "quantity", "Population"]
    ]

solar_final["capacity_per_person"] = (
    solar_final["capacity"] / solar_final["Population"]
)

solar_final["installs_per_person"] = (
    solar_final["quantity"] / solar_final["Population"]
)

solar_final["capacity_per_100k"] = solar_final["capacity_per_person"] * 100000
solar_final["installs_per_100k"] = solar_final["installs_per_person"] * 100000

solar_final
```

```
[244]:
```

	State_Full	State/Territory	Year	capacity	\
0	Australian Capital Territory	ACT	2011.0	67875.753	
1	Australian Capital Territory	ACT	2012.0	41189.284	
2	Australian Capital Territory	ACT	2013.0	52857.500	
3	Australian Capital Territory	ACT	2014.0	52175.184	

4	Australian Capital Territory	ACT	2015.0	58740.550
..	...	...	...	...
115	Western Australia	WA	2021.0	353317.798
116	Western Australia	WA	2022.0	286828.657
117	Western Australia	WA	2023.0	273494.216
118	Western Australia	WA	2024.0	285067.544
119	Western Australia	WA	2025.0	132369.570

	quantity	Population	capacity_per_person	installs_per_person	\
0	24774	367985	0.184452	0.067323	
1	13426	376539	0.109389	0.035656	
2	12556	383257	0.137917	0.032761	
3	10783	388799	0.134196	0.027734	
4	10277	395813	0.148405	0.025964	
..	...	...	...	...	
115	48780	2749365	0.128509	0.017742	
116	38691	2793934	0.102661	0.013848	
117	35213	2890936	0.094604	0.01218	
118	35535	2978147	0.09572	0.011932	
119	16450	3044382	0.04348	0.005403	

	capacity_per_100k	installs_per_100k
0	18445.249942	6732.339633
1	10938.915756	3565.633308
2	13791.659383	3276.130638
3	13419.577725	2773.412483
4	14840.480227	2596.428111
..	...	...
115	12850.887314	1774.227867
116	10266.121426	1384.821546
117	9460.403689	1218.048411
118	9571.976937	1193.191605
119	4347.994765	540.33955

[120 rows x 10 columns]

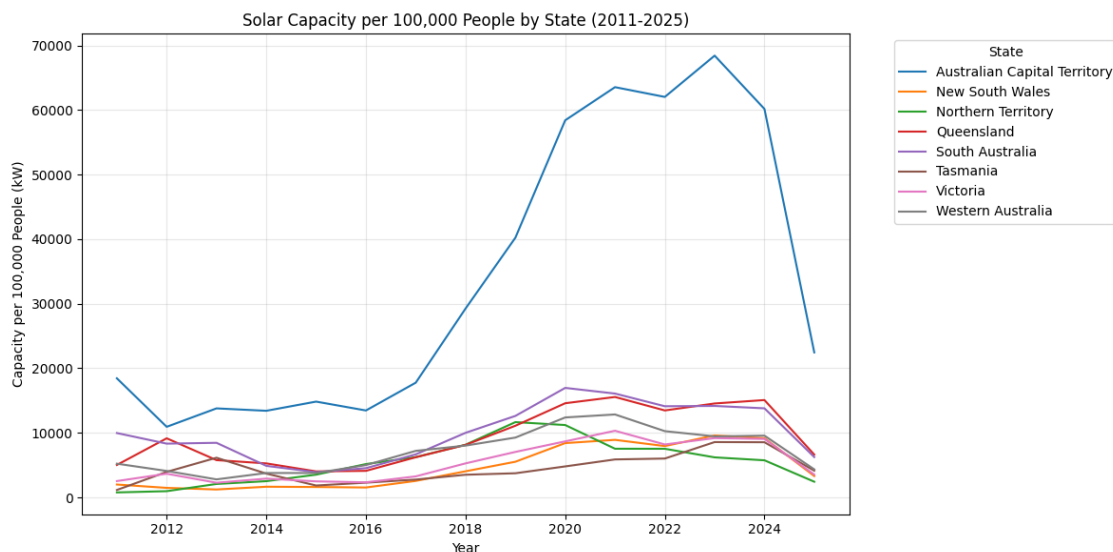
15 Visual Graph - Solar Capacity per 100k People

```
[248]: import matplotlib.pyplot as plt

plt.figure(figsize=(12, 6))

for state in solar_final["State_Full"].unique():
    subset = solar_final[solar_final["State_Full"] == state]
    plt.plot(subset["Year"], subset["capacity_per_100k"], label=state)
```

```
plt.title("Solar Capacity per 100,000 People by State (2011-2025)")
plt.xlabel("Year")
plt.ylabel("Capacity per 100,000 People (kW)")
plt.legend(title="State",bbox_to_anchor=(1.05, 1), loc="upper left")
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



15.0.1 Insights: Solar Capacity per 100,000 People

South Australia leads in capacity per 100k people, reflecting its long-standing policy support and high household solar capacity.

New South Wales and Victoria, despite having the largest populations, show lower per-capita capacity, suggesting that high population dilutes their totals when standardised.

Australian Capital Territory displays the largest peaks and drops in per-100k metrics due to its small population, which amplifies even modest changes in solar installations and capacity

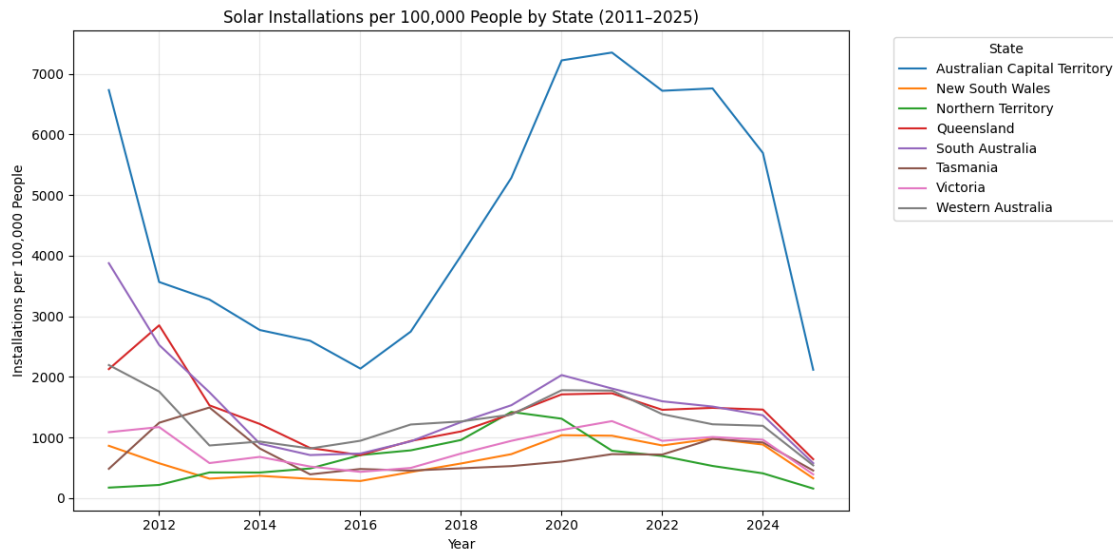
16 Visual Graph - Installations per 100k People

```
[254]: plt.figure(figsize=(12, 6))

for state in solar_final["State_Full"].unique():
    subset = solar_final[solar_final["State_Full"] == state]
    plt.plot(subset["Year"], subset["installs_per_100k"], label=state)

plt.title("Solar Installations per 100,000 People by State (2011-2025)")
plt.xlabel("Year")
```

```
plt.ylabel("Installations per 100,000 People")
plt.legend(title="State", bbox_to_anchor=(1.05, 1), loc="upper left")
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



16.0.1 Insights: Solar Installations per 100,000 People (2011–2025)

The ACT shows the most dramatic peaks and drops across the entire period. Because the ACT has a small population, even modest changes in installation numbers translate into very large per-100k values.

Queensland and South Australia consistently show high installation rates, reflecting strong household uptake, favourable solar conditions.

Conclusion This analysis examined solar capacity, installation activity, and population trends across Australian states and territories from 2011 to 2025.

The results indicate that:

- Solar adoption has increased substantially over time,
- South Australia achieved the strongest per-capita solar performance,
- Queensland maintained consistently high installation rates,
- Population-standardised metrics provide clearer comparisons between states.

Overall, the study highlights the continued expansion of solar energy adoption across Australia and demonstrates the importance of using per-capita measures when comparing regions with different population sizes.

[]: